

Course: ENCN423/627 – Sustainable Energy Technologies

Module: Battery Energy Storage Systems (BESS)

Practice problems: Battery structure/processes, Battery choice, applications

Question 1.

The two types of cell ageing (decrease in capacity) are calendar and cycling, give two examples of each

Question 2.

Which of the following is the key difference between lead-acid and lithium-ion batteries?

- a) Lead-acid batteries must be in constant use
- b) Lead-acid batteries cannot be recharged
- c) In a lead-acid battery, the anode dissolves, while in a lithium-ion battery it does not
- d) In a lithium-ion battery, the anode dissolves, while in a lead-acid battery it does not

Question 3.

What is the difference between a primary and secondary cell?

Question 4.

One of the applications of BESS in power systems is frequency regulation, describe what this is and what may happen if a power system does not have adequate frequency modulation

Question 5.

What additional component(s) can (typically) be found in

- a) A battery when compared to a cell
- b) A module when compared to a battery

Question 6.

Compare and contrast the strengths and weaknesses of BESS vs. pumped hydro storage technologies

Question 7.

In the context of a power grid, what is energy arbitrage, why is it needed, and how do BESS fill this role in renewable-heavy grids

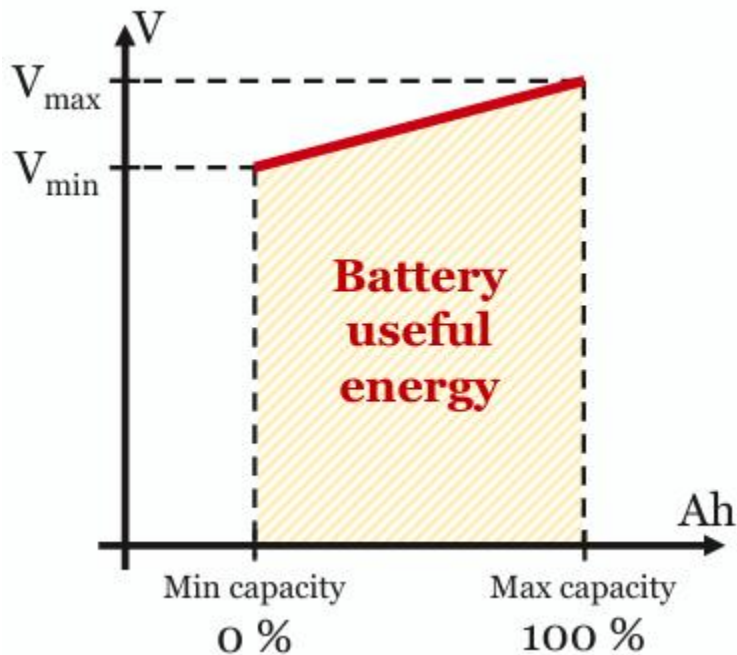
Question 8.

Describe the advantages of sodium-ion batteries when compared to lithium-ion

Question 9.

Consider a battery with a minimum operational voltage of 250 V, a maximum operational voltage of 315 V, and a capacity of 350 Ah, calculate the amount of useful energy in the battery

Hint: the following plot will be conceptually useful for this and upcoming questions



Question 10.

How would the useful energy of the battery in Question 9 change if a battery controller is applied to maintain an SoC between 10% and 80%?